FORCES

Forces can cause the following changes to objects they act on. They can:

- cause an object to accelerate—that is, to increase its velocity.
- cause an object to decelerate—that is, to decrease its velocity.
- cause an object to change its direction.
- cause a change of shape.

In order for a force to do these things, two conditions must be met.

1. ***The force must be unbalanced.***

   In order for a force to cause a change of velocity, direction or shape there must be an unbalanced force in the direction of the change.

   If the force is balanced by another force, the forces cancel each other out because they are acting in opposite directions.

This box will experience a change in its motion because the forces acting are not balanced. The larger force to the left will cause the box to accelerate to the left.

Fig 2.2.2

2. ***The force must be external.***

   Can you move a car by sitting inside it and pushing on the steering wheel?
   Of course, the answer is no.

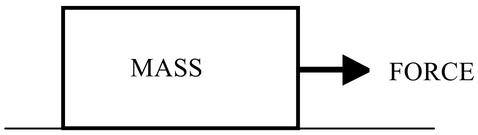
   If you wanted to push a car, you would have to stand outside the car and push it.

To fully understand how forces can do these things, one needs to understand Newton's Three Laws of Motion.

NEWTON'S FIRST LAW

> A body at rest or moving at constant velocity will continue to do so unless acted upon by an external unbalanced force.

Example



The force needed to move an object appears to be dependent on the mass.

The larger the mass, the greater the force required to move the object.

Newton's First Law is often referred to as the law of ***inertia***.

Inertia is considered to be the property of a body which, when still, makes it hard to move and, when moving, keeps it going.

This property seems to be determined by mass - ***the greater the mass, the greater the inertia***.

Examples

1. A train has a lot of inertia due to its large size. It takes a great deal of force to start it moving (if it's stationary) or stop it (if it's moving).

2. Consider a passenger in a car moving at 60.0 kmh^{-1}. If the car brakes suddenly, the passenger will continue to move forward at 60.0 kmh^{-1} (due to inertia), until the seatbelt exerts a force on them to slow them down.

Fig 2.2.6 A passenger in a car demonstrates Newton's First Law when the car slows down rapidly.

3. If a car accelerates quickly from a standing start, a passenger is thrown back in their seat. They want to stay stationary (due to inertia) and the car moves forwards under them.

When a car accelerates you feel as if you are being pushed back into your seat.

Fig 2.2.8

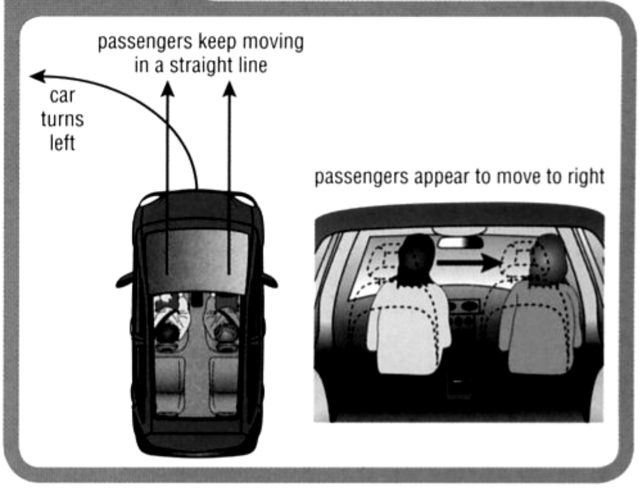
4. When a car turns a corner, the passengers have inertia that wants to keep them moving straight ahead. The car moves under them and the passengers feel as if they are moving (leaning) in the opposite direction to the turn.

   Because of this, people assume a ***centrifugal force*** is acting on the passengers pushing them away from the corner.

   THIS IS FALSE - THE "FORCE" DOESN'T EXIST!!

Fig 2.2.9

You feel as though you are being pushed to the right as the car turns left.



MORE ABOUT FORCES

Unbalanced Forces

In the diagram at right, the bike experiences an unbalanced force of 20 newtons in a forward direction.

The bike will therefore accelerate in a forward direction because of the effect of this unbalanced forward force.

To calculate the resultant force, ***take forwards as positive***.

=> The force backwards is negative.

∴ <u>ΣF = 50 - 30 = 20 N forwards</u>.

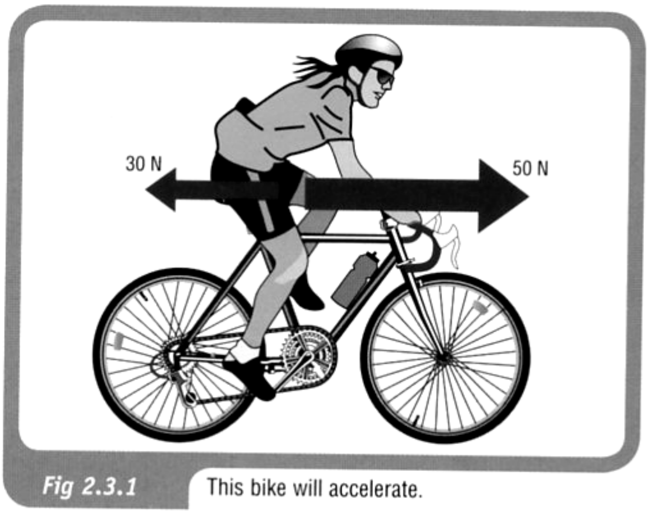


Fig 2.3.1 This bike will accelerate.

However, if the girl is not pedalling hard enough to overcome all the friction, the unbalanced force acting on the bike is 10 newtons in the opposite direction from the direction that the bike is moving.

The bike will therefore not be able to maintain a constant velocity but will decelerate.

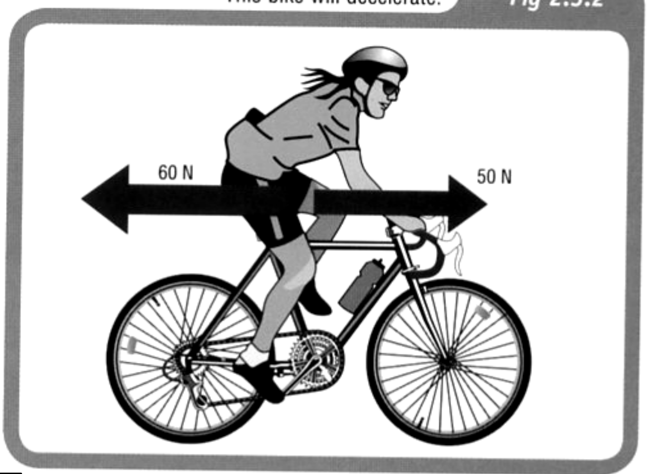
Take forwards as positive.

ΣF = 50 - 60 = -10 N

the negative sign means the direction is backwards.

∴ <u>ΣF = 10 N backwards.</u>

This bike will decelerate. Fig 2.3.2

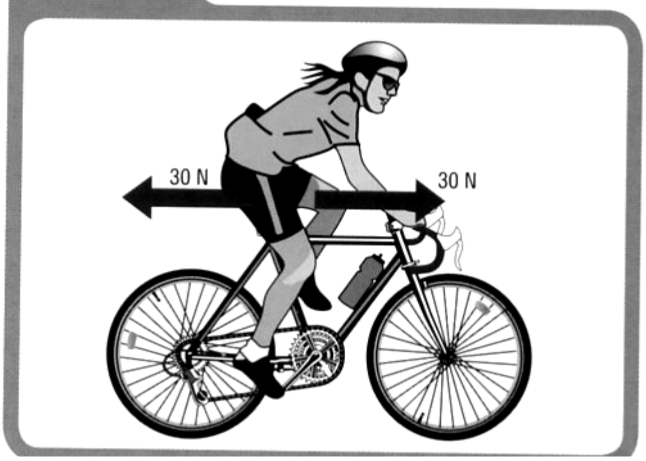


If the girl's forward pedalling force is exactly the same size as the frictional force in the opposite direction, there is no unbalanced force acting on the bike.

The bike will not accelerate or decelerate but will continue in a straight line with a constant speed.

∴ <u>ΣF = 0</u>

Fig 2.3.3 This bike will continue moving in a straight line at a constant speed.



NEWTON'S SECOND LAW

An external unbalanced force will cause an object to accelerate in the direction of the resultant force.

This is represented by:

$$\boxed{\mathbf{\Sigma F = ma}}$$

where ΣF = the resultant unbalanced force acting (N)
m = mass of the object (kg)
a = acceleration (ms^{-2})

Examples

1. What would be the acceleration of a skateboarder and her skateboard if their combined mass of 55.0 kg was acted on by an accelerating force of 60.0 N?

Take forwards as positive.

$\Sigma F = 60.0$ N

$m = 55.0$ kg

$a = ?$

$$\Sigma F = ma$$

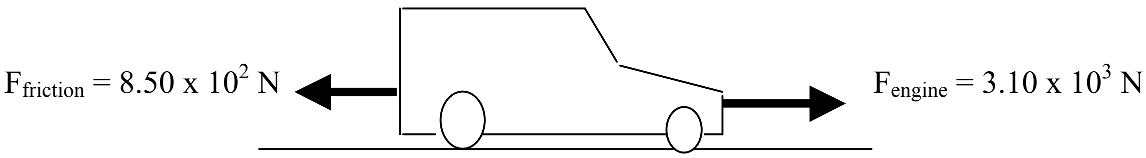
$$\Rightarrow a = \frac{\Sigma F}{m}$$

$$= \frac{60.0}{55.0}$$

$$= 1.091 \text{ ms}^{-2}$$

$\therefore$ Acceleration = 1.09 ms^{-2} forwards

2. A car of mass 1.10×10^3 kg is accelerated by a motor that provides a force of 3.10×10^3 N. A frictional force of 8.50×10^2 N opposes the motion of the car. Calculate the acceleration of the car across the ground.



Take forwards as positive.

$$\Sigma F = 3.10 \times 10^3 - 8.50 \times 10^2$$
$$= 2.25 \times 10^3 \text{ N}$$

$\Sigma F = 2.25 \times 10^3$ N

$m = 1.10 \times 10^3$ kg

$a = ?$

$$\Sigma F = ma$$
$$\Rightarrow a = \frac{\Sigma F}{m}$$
$$= \frac{2.25 \times 10^3}{1.10 \times 10^3}$$
$$= 2.045 \text{ ms}^{-2}$$

$\therefore$ Acceleration = 2.04 ms^{-2} forwards

Gravity and Weight

Most objects fall back to Earth due to gravity, one of the fundamental forces in the universe.

According to Isaac Newton, gravity is due to the attraction between two masses.

i.e. There is an attraction between you and the person next to you, your pens and pencils, the Earth, etc.

The attraction to the Earth is the strongest since its mass is so large (5.98×10^{24} kg).

Two terms need to be clearly distinguished.

Mass - amount of matter in an object (measured in kg).

Weight - force due to gravity on an object (measured in Newtons - N).

The mass of an object does not change throughout the universe while the weight will constantly change as it is moved from place to place.

As seen previously, the acceleration due to gravity on the earth's surface is:

$$g = 9.80 \text{ ms}^{-2}$$

The force due to weight (F_w) is given by:

$$\boxed{\mathbf{F_w = mg}}$$

where $\quad g$ = the acceleration due to gravity (ms^{-2})

Example

The gravitational field on Jupiter is 2.50 times greater than that on Earth. If it was possible for a person of 80.0 kg to stand on Jupiter's surface, calculate their weight.

$$F_w = ?$$
$$m = 80.0 \text{ kg}$$
$$g = (2.50 \times 9.80)\text{ms}^{-2}$$

$$\begin{aligned} F_w &= mg \\ &= (80.0)(2.50 \times 9.80) \\ &= \frac{60.0}{55.0} \\ &= 1.960 \times 10^3 \text{ N} \end{aligned}$$

$$\therefore \underline{\text{Weight} = 1.96 \times 10^3 \text{ N}}$$

NEWTON'S THIRD LAW

For every action there is an equal and opposite reaction.

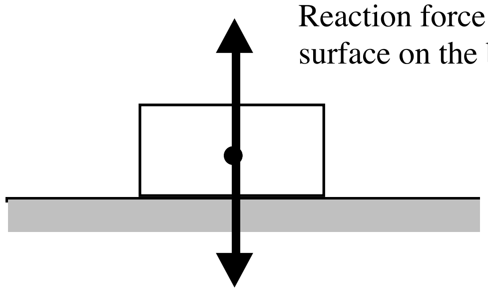
Newton realised that forces occur in pairs and act on different objects.

e.g. Consider a block sitting at rest on a table.

The force due to gravity (weight) acts downwards on the block. Thus the block exerts a force onto the table.

The table exerts an equal and opposite force to support it. The table exerts a force onto the block.

Reaction force of the surface on the block.

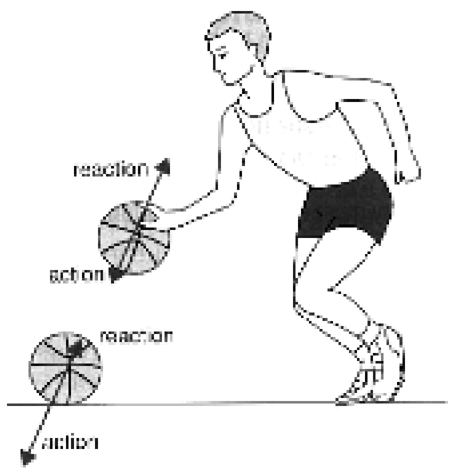


Action force of weight on the

Examples

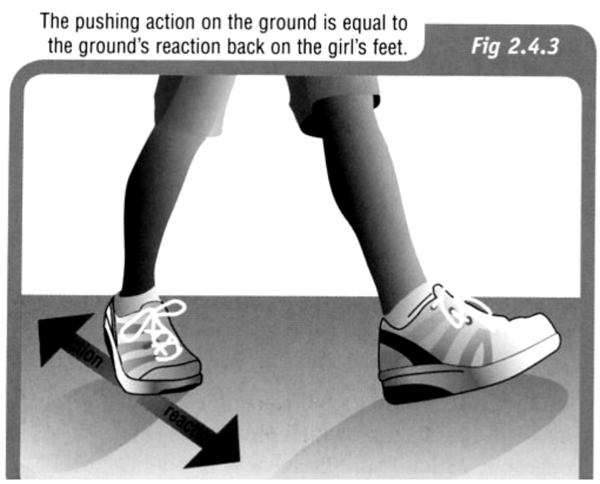
1. The player pushes on the ball (action) and the ball pushes back equally on his hand (reaction).

   When the ball hits the ground, the ball pushes on the ground (action) while the ground pushes back equally (reaction).



2. The girl is able to walk because friction allows her to push backwards on the ground.

   The ground in turn pushes back on her feet, by the same force but in the opposite direction.



3. The diagram shows the reaction chamber for a rocket.

   As the fuel burns, the expanding gases push in all directions.

   The gases rush out the exhaust outlet (action), causing a push on the rocket (reaction) that moves it forwards.

   This unbalanced force is called ***thrust***.

Fig 2.4.8 The exhaust leaving the rocket creates an unbalanced forwards force in the rocket.